

By substitution we have,

$$P_2 = P_1 + \Delta P_2 + \varepsilon_2 = (P_0 + \Delta P_1 + \varepsilon_1) + \Delta P_2 + \varepsilon_2 = P_0 + \Delta P_1 + \Delta P_2 + \varepsilon_1 + \varepsilon_2$$

This can also be written as:

$$P_2 = P_0 + \sum_{j=1}^2 \Delta P_j + \sum_{j=1}^2 \varepsilon_j$$

The discrete random walk model can then be generalized to determine the expected price P_k at any period of time k as follows:

$$P_k = P_0 + \sum_{j=1}^k \Delta P_j + \sum_{j=1}^k \varepsilon_j$$

In practice, we often make assumptions about the properties and distribution of the price drift ΔP_j and volatility ε_j terms such as a constant drift term or constant volatility.

In the case where there is no price drift term (e.g., no stock alpha over the period), the discrete random walk model simplifies to:

$$P_k = P_0 + \sum_{j=1}^k \varepsilon_j$$

Random Walk with Market Impact (No price drift)

Now let us consider the discrete random walk model without price drift but with impact cost.

Let, P_0 = arrival price, $f(x_k)$ = temporary impact and $g(x_k)$ = permanent impact from x_k shares, and ε is random noise.

The first trade price is:

$$P_1 = P_0 + f(x_1) + g(x_1) + \varepsilon_1$$

The second trade price is the equal to the first trade price plus temporary and permanent impact caused by trading x_2 shares less the quantity of temporary impact from the first trade that has dissipated from the market price at the time of the second trade. This is:

$$P_2 = P_1 + f(x_2) + g(x_2) - \{f(x_1) \cdot (1 - e^{-\gamma_1})\} + \varepsilon_2$$